

MDAnalysis Dashboard - Mini Workshop

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Google
Summer of Code



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- Pursuing BS in Data Science and Applications
 - Indian Institute of Technology, Madras
- Google Summer of Code Contributor to MDAnalysis org
 - [2025] [Better interfacing of Blender and MDAnalysis](#)
 - [2026] [Dashboard for tracking MD simulation progress with the new streaming interface](#)

Agenda



1. Streaming - Overview (5 min)
2. **mdadash** - Introduction (5 min)
3. Features (15 min)
4. Demo (15 min)
5. Interactive Activities (45 min)
6. Future Work (5 min)
7. Q & A (15 min)



Why Streaming of MD trajectories?

- “On-the-fly” Analysis of Trajectories

- **Analyze fast** (sub-picosecond) **processes** w/o writing large trajectory output files

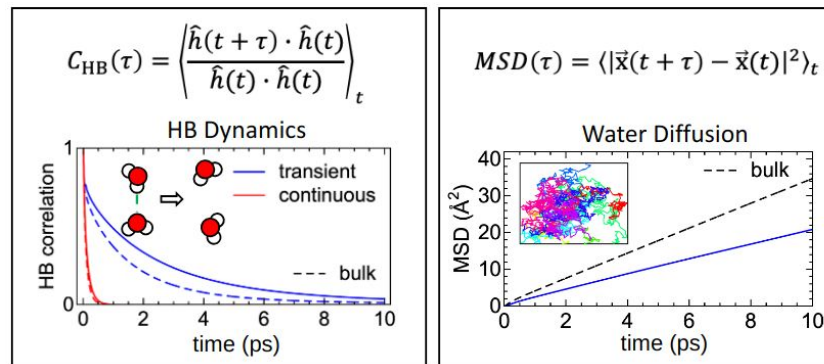
- Solvent Dynamics (HBs, diffusion)
- Molecular Vibrations
- Friction (force time correlations)

- **Event Detection**

- Monitor arbitrary collective variables
- Weighted Ensemble methods
- Water / ion flux through membrane

- **Control Output**

- Save disk space & post processing
 - Store only data you need (Eg: unwrapped protein coordinates)

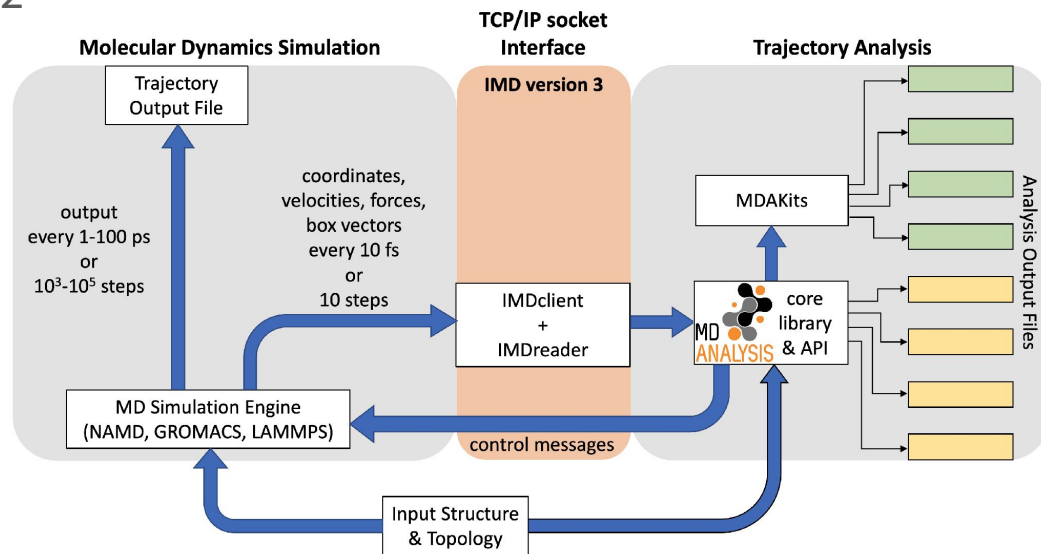


Streaming - Overview



IMDv3

- Interactive Molecular Dynamics (IMD) - [version 3 protocol](#)
- Backward compatible with IMDv2
- Implemented in:
 - **GROMACS, NAMD3, LAMMPS**
- [imdclient](#) library
- [IMDReader](#) in MDAnalysis

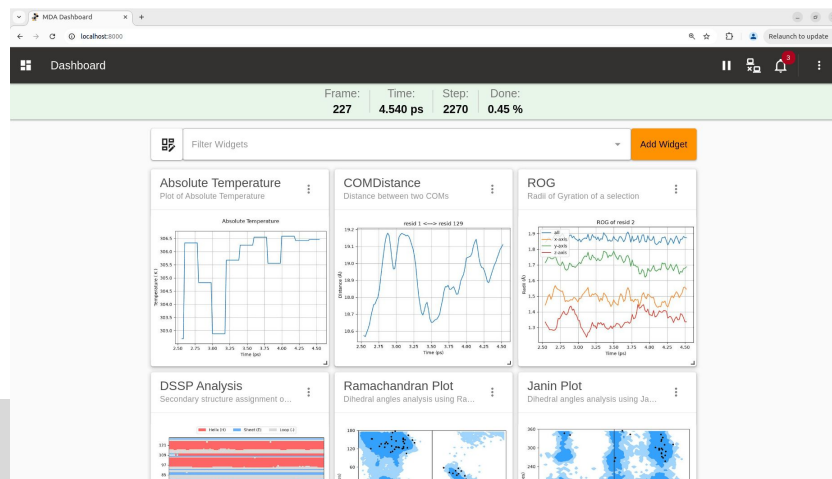


mdadash - MDAnalysis Dashboard



- Python command line tool
 - Available on PyPI and conda-forge, runs in any Python environment
- Provides browser-based real-time dashboard for monitoring, tracking and analyzing running MD simulations

```
mdadash --topology topology.tpr --trajectory imd://localhost:8889
```





- An [MDAnalysis](#) sub-project
- Uses [IMDReader](#) introduced in MDAnalysis Release 2.10.0
 - Uses [imdclient](#) to support [IMDv3 Protocol](#) (streaming)
 - [Streaming Molecular Dynamics Simulation Data for On-the-fly Processing and Analysis](#)
- Also supports regular file based trajectories
 - Considers them as running MD simulations iterating forward
 - Uses other MDAnalysis [Trajectory Readers](#) based on format
- Started as a Google Summer of Code (GSoC) 2026 [project](#)

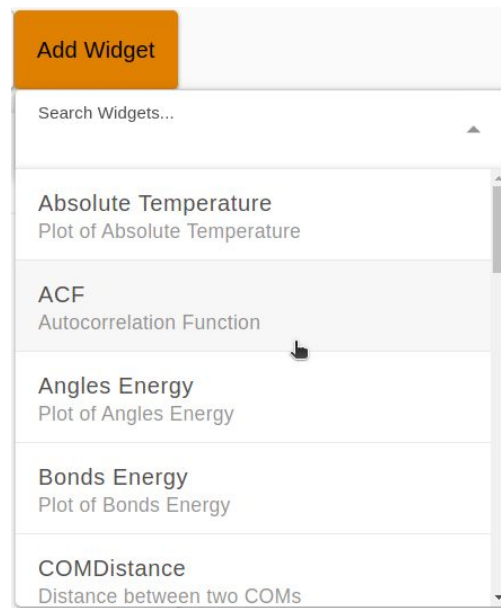
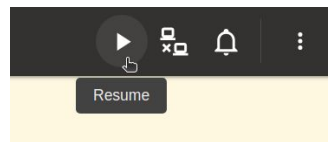


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mdadash - Features



- Control simulation (Pause / Resume / Connect / Disconnect)
- Built-in Analysis Widgets
- Buffered Access and Batching
- Parallelization
- Alerts
- Persistence
 - Single JSON based state file
- Customization
 - Custom code widget
 - Custom Analysis Widgets
 - Integrated Notebook interface
- Scalable Architecture





Built-in Analysis Widgets

- Autocorrelation Function (ACF)
 - Velocity, Position, Force
- MSD
- RMSD
- DSSP
- Simulation Energies
- Native Contacts
- Dihedral angles analysis plots
- Custom user-defined code
- ... and more
- See [full list](#) in documentation

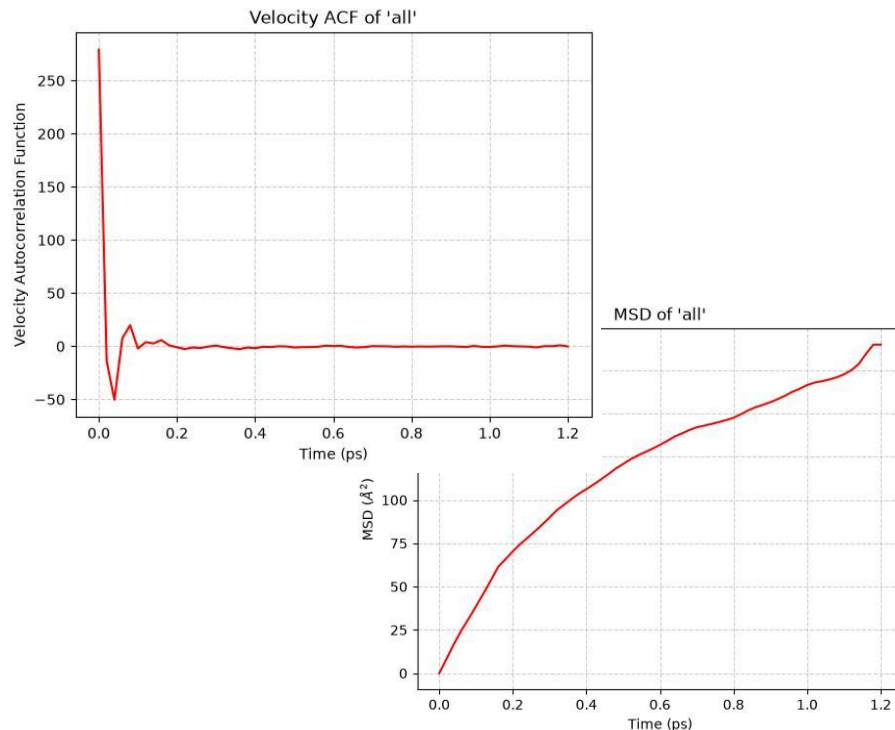
Widget	Description	Batching	Parallel
<code>acf</code>	Autocorrelation Function (ACF)	—	✓
<code>com_distance</code>	Distance between two center-of-masses	✓	✓
<code>contacts</code>	Contacts within a cutoff	✓	✓
<code>custom_code</code>	Custom user-defined code	✓	—
<code>dssp</code>	DSSP Analysis	✓	✓
<code>energies</code>	Widgets for various simulation energies	✓	—
<code>helix_analysis</code>	Helix Analysis	✓	✓
<code>hydrogen_bonds</code>	Number of Hydrogen bonds	✓	✓
<code>janin</code>	Janin plot (Dihedral angles analysis)	—	—
<code>msd</code>	MSD Analysis	—	✓
<code>native_contacts</code>	Native Contacts Analysis	✓	✓
<code>ramachandran</code>	Ramachandran plot (Dihedral angles analysis)	—	—
<code>rmsd</code>	RMSD Analysis	✓	✓
<code>rog</code>	Radii of Gyration	✓	✓

✓ Supported, ✗ Not supported, — Not applicable



Buffered Access and Batching

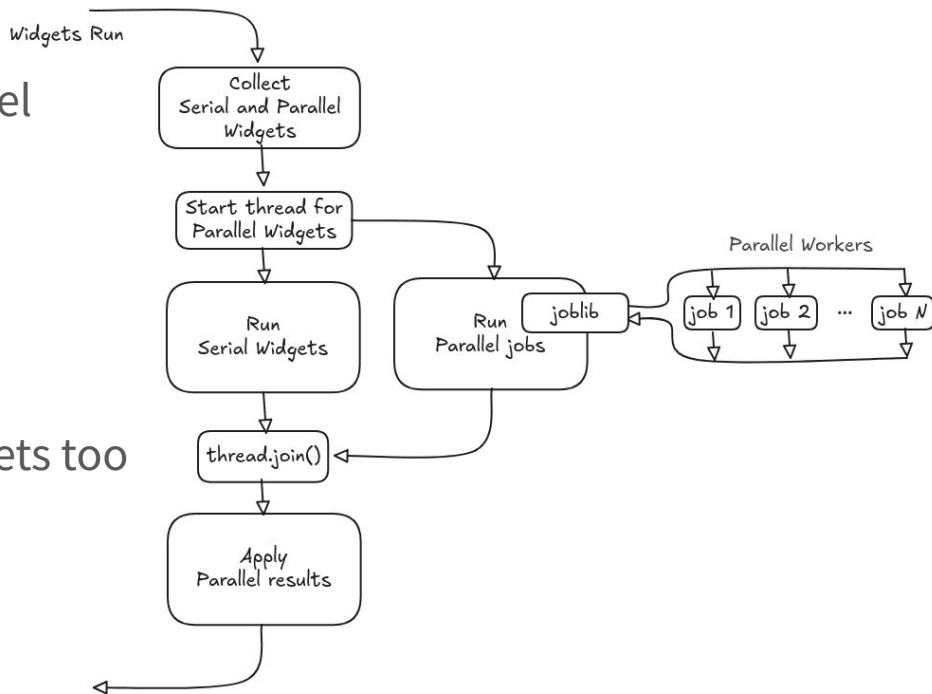
- Buffered access to last N timesteps
- Sliding window
- Allows for lag-time analyses
- Allows batch run of analyses
- Supports [AnalysisBase](#)-based classes
 - Not possible with regular [IMDReader](#)
 - Trajectory length equals buffer size
- See [Batching](#) in documentation





Parallelization

- Most built-in widgets can run in parallel
- Process based parallelization
 - Analyses CPU-bound
 - Uses [Joblib](#)
 - Setting to control max jobs / processes
- Can be combined with Batching
- Is supported in Custom Analysis Widgets too
- See [Parallelization](#) in documentation





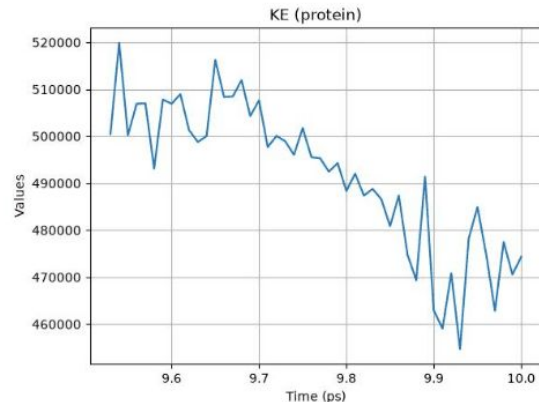
Custom Code Widget

- Custom user-defined code
 - Notebook cell interface
- Setup code
 - Run once during widget creation
- Execute code
 - Run as per the widget run frequency
- A quick way to run simple custom code
- See [Custom Code](#) in documentation

```
^ Execute code ▶  
1 ke1 = ke(u.select_atoms("protein"))  
2 print(f"Protein KE = {ke1:.2f}")  
3 plt1.show("KE (protein)", ke1)  
  
This code will run as per the run frequency
```

Output

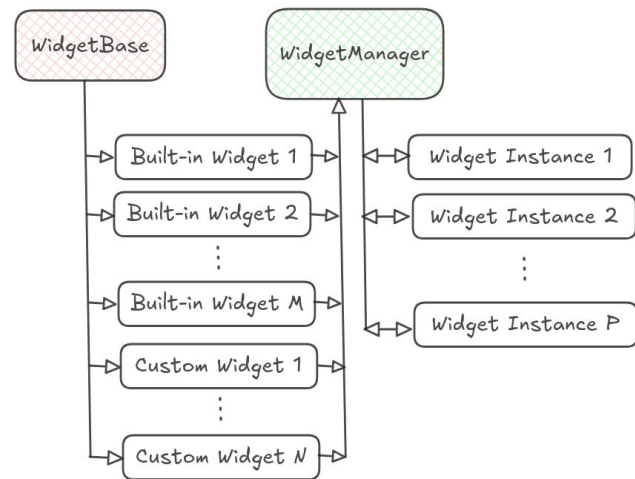
Protein KE = 474323.49





Custom Analysis Widgets

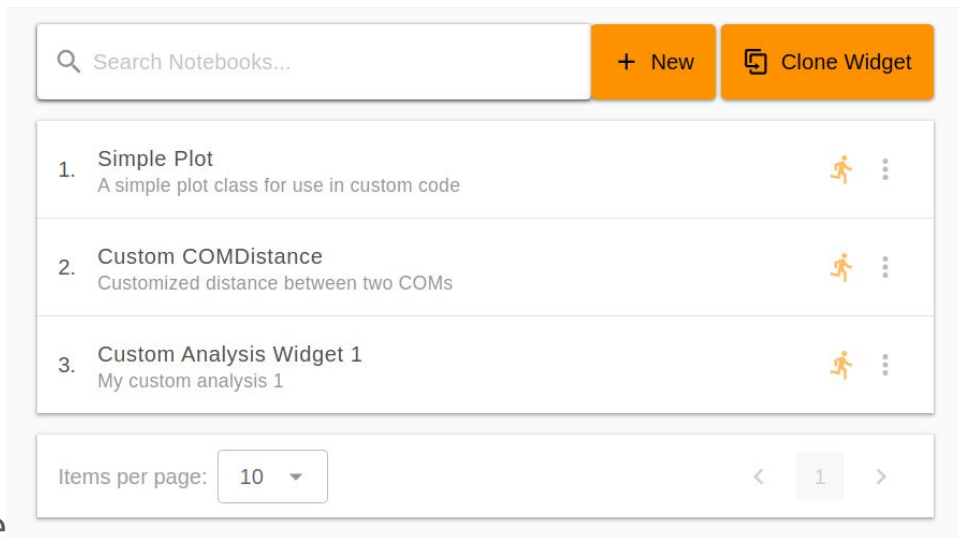
- Single common framework for Widgets
 - Same used by Built-in Analysis Widgets
 - All derive from `WidgetBase` base class
- Support for different run frequencies
 - Every-frame or Batch
- Support for different run modes
 - Serial or Parallel
- Dynamic inputs and lifecycle hooks
- See [Adding Custom Widgets](#) in documentation





Integrated Notebook Interface

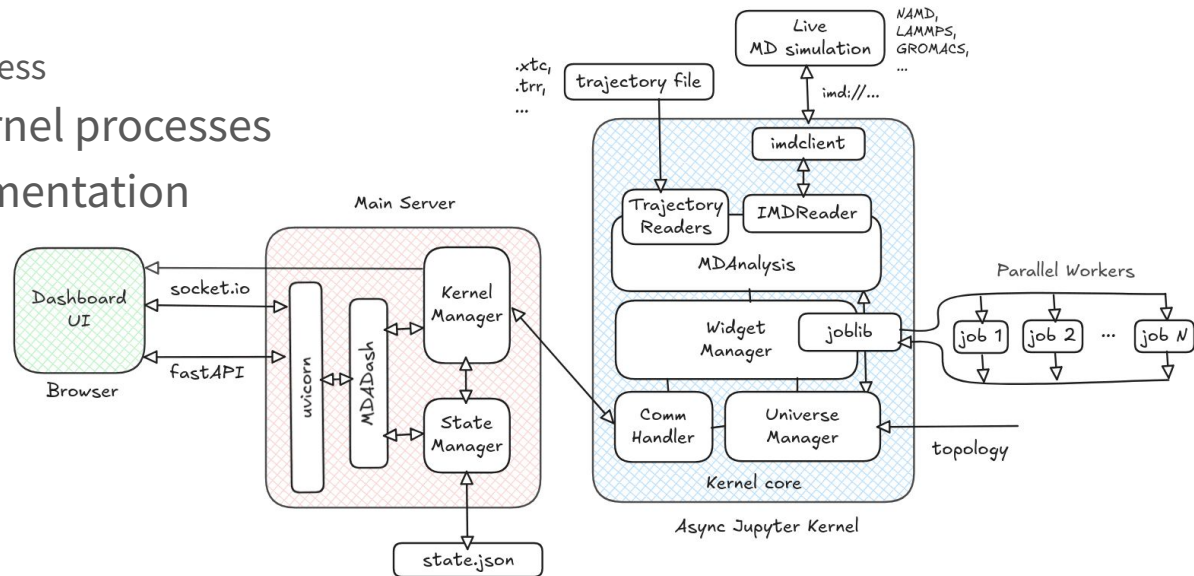
- Custom functions and Classes
 - Utils, custom plots, etc
- Notebook cells
 - Code complete support
 - Execute, re-order, delete, etc
- Run on Launch support
 - Setup custom functions / classes
 - Register custom widgets
- Clone Built-in Analysis Widget code
- Automatic refresh of widget instances





Scalable Architecture

- Two processes
 - Main Server process
 - Async Jupyter Kernel process
- Extensible to multiple kernel processes
- See [Architecture](#) in documentation



Demo and Activities



- Goto repo: <https://github.com/PardhavMaradani/mdadash-mini-workshop-2026>
- Follow instructions to “Open in GitHub Codespaces”
- Follow “activities.md”

The screenshot shows the README file for the repository 'github.com/PardhavMaradani/mdadash-mini-workshop-2026'. The title is 'MDAnalysis Dashboard - Mini Workshop'. The text welcomes users to the 'mdadash' mini workshop and provides setup instructions. It explains how to create a GitHub codespace by clicking a button labeled 'Open in GitHub Codespaces'. A note mentions that if the preview browser doesn't show up, users can click on the 'Ports' tab of the bottom panel and follow the link for the 'mdadash server port' to open the dashboard. Finally, it lists two ways to stop or delete the codespace: clicking the 'Code' button and switching to the 'Codespaces' tab, or going to the 'GitHub Codespaces' page and selecting the codespace to delete.

github.com/PardhavMaradani/mdadash-mini-workshop-2026

README

MDAnalysis Dashboard - Mini Workshop

Welcome to the [mdadash](#) mini workshop.

Setup instructions

GitHub Codespaces

You can create a GitHub codespace by clicking on the badge below (Use **Ctrl / Cmd + Click** to open in new tab. A **4-core** 'Machine type' is recommended):

 Open in GitHub Codespaces

Once the codespace is fully setup, it will automatically launch a preview browser window showing the dashboard in the editor area.

Note: If the preview browser doesn't show up or if you closed it, you can click on the **Ports** tab of the bottom panel and follow the link for the 'mdadash server port' to open the dashboard.

You can stop / delete the Codespace after use in either of the following ways:

- Click the "Code" button in this repo, switch to the "Codespaces" tab and select the codespace to stop / delete
- Go to your [GitHub Codespaces](#) and select the codespace to stop / delete



- Real-time Visualization
 - [Mol*](#) support in browser
 - [MolecularNodes](#) support on desktop
- Extend Notebook functionality
 - Support for Markdown, Import / Export, etc
- More Built-in Analysis Widgets
- Ensemble of simulations
- Feedback from the community

Q & A

- GitHub Repo: <https://github.com/MDAnalysis/mdadash>
- Documentation: <https://mdadash.readthedocs.io/>



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